

COMPSCI 4NL3: Natural Language Processing

Syllabus Winter 2025

1 Course Description

Artificial intelligence has exploded over the past decade and the processing of natural language has been a central part of this growth. This course will cover the fundamentals of the field of natural language processing while focusing on recent developments primarily related to the use of deep neural networks and how they have enabled large language models.

This course will not go in-depth into machine learning algorithms but will discuss essential NLP architectures and machine learning basics. Assignments and lectures will focus on the Python language and will require programming background. Overview of natural language processing with a focus on recent developments: Basics of data processing and machine learning in the context of NLP, text classification and sequence labeling, vector semantics, embeddings, language modeling, translation, question answering, parsing, dialogue and conversational systems, discourse processing, and large language models. Practices and developments will be discussed with practical exercises focusing more heavily on recent neural network approaches.

Three lectures, one tutorial (one hour); one term

Prerequisites:

1. One of COMPSCI 2C03 or SFWRENG 2C03 or MECHTRON 2MD3
2. One of MATH 1B03 or 1ZC3
3. One of STATS 2D03 or 3Y03

Lecture: Tuesday, Wednesday, Friday 9:30am - 10:20am

Tutorial: Tuesday 10:30am - 11:20am and Friday 11:30am - 12:20pm

2 Instructors

Professor Welch is available by appointment. If you have questions about the class and material and your question does not contain personal information it must be posted on the teams channel. We will monitor the channels during work hours.

Professor: Charles (Charlie) Welch

TA: Amirhossein (Amir) Sabour and Xiaochen (Chris) Li

3 Books

The book that will be most useful for this course is the first one. This is the go-to book for natural language processing. The latter two books will teach you more about the modern foundations of deep learning with the former being more focused on natural language processing and the latter being more generally informative about the topic of deep learning. These books are free online. You do not have to purchase anything.

1. Dan Jurafsky and James H. Martin. Speech and Language Processing. <https://web.stanford.edu/~jurafsky/slp3/>
2. Yoav Goldberg. A Primer on Neural Network Models for Natural Language Processing. <https://arxiv.org/pdf/1510.00726>
3. Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. <https://www.deeplearningbook.org/>

4 Tentative Schedule

The lectures for the course cover different topics each week. This is subject to change.

- W1: Intro to Natural Language Processing: Text processing basics, regular expressions
- W2: Supervised Text Classification: Logistic regression, naive Bayes, introduction to neural networks
- W3: Language Models: N-gram models, neural models, machine translation
- W4: Word Vectors: Latent semantic analysis, word2vec, GloVe, contextual embeddings, morphology
- W5: Sequence Labeling: Hidden markov models, part-of-speech tagging, LSTM networks
- W6: Syntax: Context-free parsing, dependency, constituency, semantic parsing
- W7: Data Collection: Annotation, crowd sourcing, bias, ground truth, subjectivity
- W8: Unsupervised Learning: Clustering, expectation maximization, topic modeling
- W9: Neural Networks: BERT, Transformers, reinforcement learning with human feedback, safety, efficiency
- W10: Discourse: Narrative, chatbots, co-reference
- W11: Lexical Semantics: Word sense disambiguation, entity linking and detection, semantic roles and frames, diachronic change
- W12: Application Areas: Sociolinguistics, social NLP, information extraction, inference, question answering

5 Learning Objectives

1. Describe linguistic foundations of Natural Language Processing
2. Apply statistical and mathematical concepts for language representation
3. Create and deploy computational text processing and analysis methods
4. Apply modern machine learning tools and frameworks in the domain of NLP
5. Build end-to-end systems which accomplish a range of NLP tasks
6. Evaluate NLP applications and their effects on users and society

6 Course Work and Grading

Your grade will be comprised of the following:

- 20% - Five programming homework assignments, each worth 4%
- 20% - Group project (4 stages, see below)
- 30% - Midterm Exam
- 30% - Final Exam

The midterm will take the place of one class period in the middle of the semester and last for 50 minutes, making it shorter than the final exam, which will occur during the final exam period. The final exam is not cumulative but will focus on material from the second half of the course.

The Python language will be required for the four assignments. The fifth assignment and group project can be done in the language of your choosing. I recommend installing Conda to manage programming environments so that you can separately manage dependencies without interfering with other things on your system. Assignments are due on their delivery date. Recognizing that, according to universal design for learning principles (UDL), some students might require more time, there is a 5-day tolerance for late delivery, with no penalty.

7 Group Projects

For the group project, you will need to find a group of 3-4. There will be a teams channel for people looking for groups. The final project will be completed in four steps.

1. Propose a dataset - In this step, your team will identify a classification task you want to complete. The data you choose must not be already labeled.
2. Develop annotation plan - Your group will develop the annotation plan for the dataset and create an annotation tool.
3. Annotate other group data - In this stage, each individual will annotate data for two other tasks
4. Report on lessons learned - Your group will write a 4 page report using the ACL-style LaTeX template describing your task, the data, annotation process, attempts to solve your task, results, and a discussion.

Separate handouts will be provided for each step of the final project at later dates.

8 Generative AI Policy

You may use generative AI to help you with your assignment and projects. However, you must follow the privacy guidelines (see next section) and you must report (1) how you used generative AI and (2) the carbon footprint of using these models. **You must disclose how you use the technology.** There is no penalty for using it but you will receive a zero on the assignment if you do not disclose how you used it.

There is a calculator you can use here to calculate the CO2 emissions. When using this, please report the (1) name of the model, (2) hardware type, (3) time used, (4) provider, (5) region of compute, (6) very brief description of how you determined these values (it might be less clear on some platforms but the point is to show you put some effort into approximating the answers). ChatGPT changes over time as they update the model but if you are using ChatGPT you can this estimate of 4.32g CO2 per search query.

9 Privacy Disclosure and Licensing

Course materials are licensed by the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License. You are not permitted to upload course materials to public commercial websites. This includes study help websites and websites that use generative AI, such as ChatGPT. Generative AI websites use these materials to further train their models. While you cannot upload the assignments or questions, you can, for instance, ask ChatGPT variations or subparts of the problem. Break down the task and if you want to use generative AI, ask it about the more specific parts you get stuck on.



10 Course Evaluation (Extra Credit)

Extra credit will be given to all students depending on the total class participation in course evaluations. They help me improve as an instructor so I appreciate you taking the time to fill them out. The following will be awarded to all students given the corresponding participation rate:

1. > 50% All individual assignments rounded up

2. > 60% All project checkpoints rounded up
3. > 70% All exams rounded up
4. > 80% Final weighted grade rounded up

11 MSAF Policy

Type-A MSAFs are automatically accepted. You may MSAF the midterm, in which case the final exam is then worth the combined amount of both exams. You may submit your MSAF directly through Mosaic. Type-B MSAFs will be discussed with the student after being accepted by the Faculty.

12 Academic Integrity and Plagiarism

You are expected to exhibit honesty and use ethical behavior in all aspects of the learning process. The academic credentials you earn are rooted in honesty and academic integrity principles.

Individual assignments must be solved by one person only, any outside source (this includes asking other people or using any books or information found on the web) has to be documented. Go to cfwelch.com/ec for extra credit. In compliance with the senate regulations on academic integrity, I remind you that: People who let other people copy are as guilty as the ones who copy. You can consult outside sources, meaning textbooks or the web, but any use of an outside source must be documented. Code plagiarism detection tools will be used systematically.

Similarly, group assignments must be solved by members of that group only, and the University policies apply as well. Students must sign an integrity pledge on Avenue to access the assignment drop boxes.

13 Advisory Statements

13.1 Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behavior can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>.

The following illustrates only three forms of academic dishonesty:

1. plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained
2. improper collaboration in group work
3. copying or using unauthorized aids in tests and examinations

13.2 Authenticity/Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

13.3 Courses with an Online Component

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

13.4 Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

13.5 Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviors that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

13.6 Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

13.7 Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review the Policy on Requests for Relief for Missed Academic Term Work.

13.8 Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations.

Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

13.9 Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

13.10 Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.